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Advances in multi- and hyperspectral remote sensing of mangrove species: A synthesis and study case on airborne and multisource spaceborne imagery

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ABSTRACT

This study summarizes the advances in mangrove species mapping based on multispectral and hyperspectral imagery achieved over the last decade. The influence of species diversity and sensor specifications on the performances of various classification approaches are discussed. Based on the limitations of previous approaches, we propose a novel framework to map mangrove species at medium, high, and very-high spatial resolution using multispectral and hyperspectral images. The framework relies on a multitask convolutional neural network to achieve accurate classification at pixel and object (*i.e.* individual tree crown) level. It was successfully applied to the most comprehensive dataset of optical imagery ever collected over a mangrove forest, achieving accurate mapping of species on airborne (OA = 95 %, Kappa = 0.93) and spaceborne imagery (OA and Kappa up to 97 % and 0.95, respectively), including the newly-operating hyperspectral DESIS and PRISMA instruments. This study brings new insights into the role of spatial and spectral resolutions in mangrove species classification, particularly the importance of short-wave infrared bands. Multispectral imagery performs well at very-high to high resolutions (up to 10–20 m) but suffers from the lack of spectral information at medium resolution (30 m). Hyperspectral imagery provided the best results at sub-metric resolution (0.5 m) and satisfactory results at 30-m resolution (OA $\geq$ 85), and could benefit from implementing spectral unmixing in the latter case. Given the increasing attention accorded to mangroves, remote sensing will undoubtedly become an essential tool to monitor the diversity of these endangered ecosystems in the future. In that respect, efforts should concentrate on evaluating the capabilities of new and upcoming instruments and multisource data combination to improve mangrove species mapping and, more broadly, to meet conservation goals.

1. Introduction

Mangroves thrive in tropical and neotropical regions of 118 countries worldwide, in the transition zone between marine and terrestrial environments (Duke, 2020; Makowski and Finkl, 2018). They cover about 13.5 million hectares of coastline and provide more than 20 ecological services (Barbier, 2016; Bunting et al., 2018; Thomas et al., 2017; Vo et al., 2012). They also act as barriers to coastal erosion and contaminant transportation and contribute to the global blue carbon pool more than any other ecosystem (Lewis et al., 2011; Nagelkerken et al., 2008; Zeng et al., 2021). Despite their benefits and importance, mangroves have been impacted by degradation and deforestation, with about 35 % of global mangrove extent lost since the late 20th century. Consequently, their conservation has gained increasing interest worldwide aiming to maintain their invaluable services (Bunting et al., 2018; Ward et al., 2016).

Mangroves are considered biodiversity hotspots due to the high diversity of terrestrial and marine species they shelter (Cannicci et al., 2021; Nagelkerken et al., 2008). Mangrove forests comprise more than 70 tree, palm, shrub, and fern species (Duke et al., 1998; Simard et al., 2019), of which up to 58 different can be found in a single stand (Rastogi et al., 2021). However, their distribution varies at global, regional, and

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local scales, as well as their sensitivity to natural and anthropogenic pressures such as sea level rise and environmental pollution (Bullock et al., 2017; Duke, 2017; Schaeffer-Novelli et al., 1990). Since habitats and resources provided by mangrove forests differ among species, their mapping directly contributes to monitoring habitat and resource availability for the associated fauna (Rog et al., 2017). More generally, the ecological services provided by mangroves vary according to their specific composition (Lee et al., 2014; Mitra, 2020), highlighting the importance of species mapping for mangrove conservation and restoration.

Remote sensing can inform on the richness, zonation, dominance, and type of species (true mangroves and associates), and their response and resilience to stressors and disturbances. For example, Lombard and Andrieu (2021) highlighted changes in species zonation in Senegalese mangroves by combining Landsat ETM + and 8 OLI imagery. Similarly, Bullock et al. (2017) analyzed Landsat time series to monitor the colonization of a Mekong shoreline by four mangrove species over decades. Focusing on disturbances, Piou et al. (2006) and Duke et al. (2017) reported modifications of species zonation following hurricanes and extreme weather events, respectively. Similar changes may also occur after major pollution events such as oil spills, as mangrove species differ in their tolerance to pollutants (Duke et al., 1999; Lewis et al., 2011).

Likewise, species mapping allows calibrating species-specific models of biophysical and biochemical trait retrieval, and biomass and blue carbon stock estimation based on remotely sensed data (Flores-de-Santiago et al., 2016; Komiyama et al., 2008; Kovacs et al., 2005; Pastor-Guzman et al., 2015; Zhang et al., 2014). Although generic models show satisfying performance in global estimates (Sanderman et al., 2018; Simard et al., 2019), species-specific models are essential to retrieve canopy traits and to detect early mangrove stress accurately. Hence, species mapping is essential for mangrove monitoring and conservation applications, including in low-diversity regions. Species maps obtained by optical imagery can be used either as an end goal, to evaluate the richness and distribution of species within a single stand or across sites, or as an intermediary product to accurately retrieve mangrove attributes (traits, biomass, *etc.*).

Although key advances have been made in mangrove species mapping in the past, no dedicated review has been proposed so far. Thus, the primary objective of this work is to summarize the performances and limitations of existing approaches in this field, considering multiple factors (sensor type, resolution, species diversity, pixel- *vs* object-oriented mapping, algorithms, *etc.*). Then, we propose a novel framework to map mangrove species based on the most comprehensive dataset of optical imagery ever collected on mangroves, including airborne and spaceborne imagery at varying spatial and spectral resolutions. This framework aims to overcome the limitations of existing approaches identified in the literature. These two components are presented alternatively and discussed jointly throughout the article.

2. Materials and methods

2.1. Literature review

We looked for published articles on mangrove species mapping by combining "MANGROVE" and "SPECIES MAPPING/CLASSIFICATION" keywords to "SATELLITE / SPACEBORNE IMAGERY", "AIRBORNE / AERIAL IMAGERY", "DRONE / UNMANNED AERIAL VEHICLE IMAGERY", "OBJECT-ORIENTED / PIXEL-ORIENTED CLASSIFICATION", "MULTISPECTRAL / HYPERSPECTRAL IMAGERY" in Scopus and Google Scholar search engines. Considering only articles based on imaging spectroscopy and rigorous field-based assessment (*i.e.* using ground truth data), we analyzed 287 studies. For each of them, we extracted the following information: image type (Unmanned Aerial Vehicle (UAV), airborne, spaceborne), instrument name (spaceborne imagery only), instrument specifications (spatial and spectral resolutions), number of mangrove species, type of mapping (pixel- *vs* object-oriented), and classification algorithms. Although comparisons of studies remained limited by the high variability in the size of ground truth datasets, some trends could still be drawn. So, we compared the performances of species mapping considering the abovementioned factors, based on the Overall Accuracy (OA) and Cohen's Kappa coefficients reported in studies. These metrics were selected because they are the most widely used for classification assessment in remote sensing. Therefore, they have been systematically reported in previous studies, unlike other metrics. However, comparisons based on OA and Kappa values should be interpreted with caution here, as these metrics are influenced by multiple factors such as class imbalance and spatial autocorrelation. Based on this review, we identified the main limitations of existing approaches, which were considered while designing our species mapping framework.

2.2. Study site

The study site selected to develop our framework is a fringe mangrove forest of 330 ha located in the São Paulo state, southeastern Brazil (Fig. 1). The forest is bordered by the Bertioga channel, a brackish river flowing into the Atlantic Ocean. Trees populating this mangrove are exposed to 1.5-m range tides, which flood the forest in its seaward segment and along river meanders reaching its inner part. Conversely, the landward segment is poorly influenced by tides and brings freshwater to the forest through small rivers. Three true mangrove species with distinct zonation patterns have established in this region owing to its hydrological context: *Rhizophora mangle* (L.) (red mangrove), *Avicennia schaueriana* Stapf & Leechm. ex Moldenke (black mangrove), and *Laguncularia racemosa* (L) (white mangrove) (Schaeffer-Novelli et al., 1990). These species exhibit distinct tree morphologies and leaf greenness, as illustrated in Fig. 2. Red mangrove usually flourishes along the forest border and river meanders as it is well-adapted to continuous



Fig. 1. Geographic location of the study site in the São Paulo (SP) state, southeastern Brazil. Mangrove areas as mapped by Bunting et al. (2018) are indicated in green. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

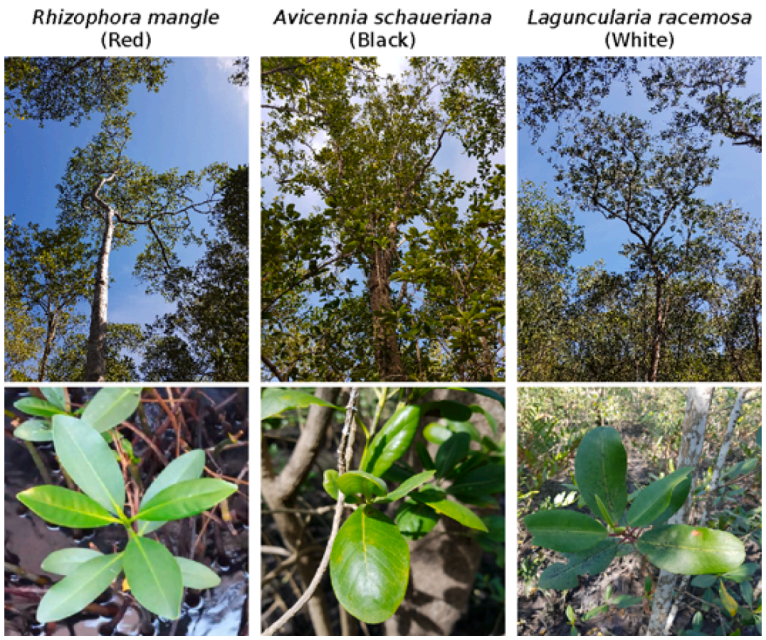


Fig. 2. Mangrove tree species found in the Bertioga forest.

flooding. Crowns of red mangrove trees are often large and loose in shape, and leaves from this species are usually deep green under favorable growing conditions. Black mangrove mainly establishes in the innermost part of the forest, which is periodically flooded. This species forms dense, compact tree crowns composed of curved, bright green leaves. The white mangrove grows in specific locations along the landward segment of the forest, which is only flooded during the highest tides. This species is less competitive than the red and black ones under saline conditions. Tree crowns of white mangroves are typically sparse and round-shaped with elliptic green leaves. Although species are spatially well-distinguished, many individuals have established outside their preferential zone. That is, individuals from the three species are sometimes found in a relatively small area, making their mapping particularly challenging.

2.3. Field survey

A field survey was carried out in the Bertioga mangrove in August 2021 to build the ground truth dataset of mangrove species. This period corresponded to the dry season, the most favorable for cloud-free conditions. Tree positions were recorded at their crown center using a Trimble TSC5 rover connected to an SP85 GNSS receiver (Trimble Inc., Sunnyvale, United States), providing a centimetric precision after post-processing in the Survey Office software® (Spectra Geospatial, Westminster, United States). For each tree, the species was identified and its coordinates were informed in a shapefile. To take the spatial heterogeneity of the forest into account, we surveyed trees located within and outside their preferential zone for each species (*e.g.* we included black mangrove trees located in the red mangrove zone). Likewise, we selected trees from varying configurations (isolated, fringe, clumped), crown shapes (round, elliptic), and ages (young, mature) to be representative of the forest. The ground truth dataset was later extended by photo-interpretation of the Very-High-Resolution (VHR) images (see Section 2.2.5) and served for training and assessing the classification models presented in this study.

2.4. Image acquisition and preprocessing

Multi- and hyperspectral images acquired by six different instruments over the Bertioga mangrove were exploited in this study. This included airborne and spaceborne images, whose specifications, and preprocessing steps are elaborated in detail below (see also Table S1 and the detailed flowchart in Figure S1). Whenever possible, the images were collected during the same period of the field survey. All images underwent the same processing pipeline, which included preprocessing, mangrove masking, ground truth dataset generation and splitting, model training and assessment, and finally prediction over the entire area (*i.e.* final species mapping at pixel and/or object level).

2.4.1. Airborne

Airborne hyperspectral images were acquired over the whole forest on August 10th, 2021 by Fototerra (Santana de Parnaíba, Brazil) from an aircraft equipped with an AisaFENIX 1 K instrument (Specim, Spectral Imaging ltd, Oulu, Finland). The AisaFENIX 1 K instrument includes modules covering the [380–2500] nm domain with 364 spectral bands at a 5–6 nm spectral resolution. The forest was covered by 25 flight lines with 15 % overlapping acquired under clear sky on a single day in a two-hour interval. The acquisition system included an Inertial Navigation System (INS) equipped with an inertial measurement unit, which combines GNSS and line-of-sight data to deliver an Internal Geometry Map (IGM) file that contains the per-raw-pixel geocoding information. IGM information served for creating a geographic look-up table comprising the locations of each pixel in each flight line. The plane was flown at an altitude of 440 m above sea level, resulting in a spatial resolution of 0.5 m. The resulting NIR-green–blue false-color mosaic of the Bertioga mangrove is illustrated in Fig. 3. Geometric correction was applied to the radiance images of each flight line using their respective geometric lookup table. Images were then mosaicked and atmospherically corrected in ENVI® (L3Harris Geospatial, Boulder, CO, USA) using the Fast Line-of-Sight Atmospheric Analysis of Spectral Hypercubes (FLAASH) module based on the MODTRAN4 radiative transfer model. The model was applied to the radiance image using the parameters described in Ferreira et al. (2016). We selected tropical atmospheric and rural aerosol models as input parameters, and the 2-Band (K-T) method was used to retrieve aerosols and estimate the scene visibility. The 1135-nm wavelength was selected to retrieve the water vapor in each pixel. Finally, spectral signatures were smoothed using a three-band moving window to improve the signal-to-noise ratio while removing spectral artifacts. Bands from the [1350:1450] and [1750:2050] nm regions were removed because of the low signal-to-noise ratio (SNR) caused by the

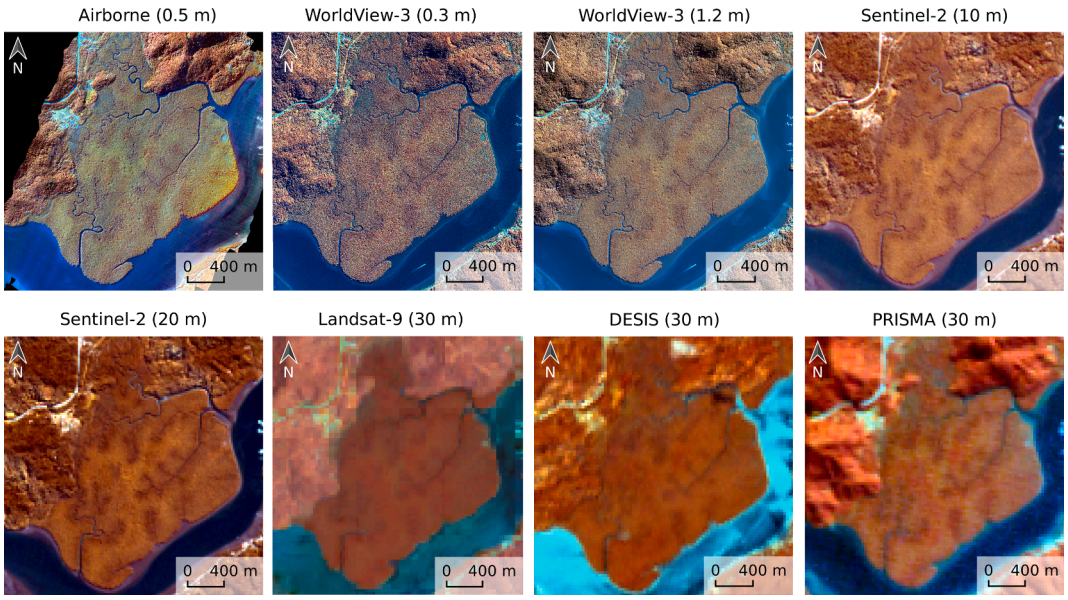


Fig. 3. Bertioga mangrove forest as seen by the six instruments exploited in this study (Near-Infrared-Green-Blue false-color composites). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

atmospheric water absorption. Also, we discarded the noisy bands below 400 and above 2400 nm. Finally, 269 bands were kept for further processing of the reflectance image.

2.4.2. *Worldview*

WorldView-3 satellite imagery was acquired on July 24th, 2021 over the study site under cloud-free conditions (Fig. 3). A single tile covered the whole forest with an off-nadir angle of 19° and a sun elevation of 39.3°. The corresponding level-2A product (*i.e.* surface reflectance) provided by MAXAR (Westminster, CO, USA) was exploited in this study. The product included a panchromatic band (PAN) at 0.3-m resolution, eight Visible-Near-Infrared (VNIR) bands at 1.2-m, and eight Short-Wave Infrared (SWIR) bands at 3.7-m. Two VHR images were generated from these products. First, the VNIR and SWIR images were pan-sharpened using the Nearest Neighbor Diffusion (NNDiffuse) algorithm implemented in ENVI® (Sun et al., 2014). NNDiffuse exploits the spectral mixture analysis concept, assuming that a pan-sharpened pixel spectrum can be modeled from the spectral response of its neighboring pixels. In a previous study, this algorithm appeared the most suited for WorldView imagery on the same site (Lassalle et al., 2022). In addition, recent studies demonstrated that pan-sharpening could be applied to both the VNIR and SWIR bands of WorldView-3 over vegetated regions without losing the spectral information of the native resolution images (Collin et al., 2018; Ferreira et al., 2019; Sagan et al., 2021). The NNDiffuse algorithm was applied with a pixel size ratio of 4 and 12 for the VNIR and SWIR bands, respectively, using a spatial smoothness of 2.48 and a dynamically-adjusted intensity smoothness. Finally, the resulting pan-sharpened VNIR and SWIR bands were stacked, resulting in a single 16-band image at 0.3-m spatial resolution. A second VHR image obtained from the native WorldView products resulted from the resampling of SWIR bands (3.7 m) to the resolution of the VNIR ones (1.2 m). Hence, two 16-band images covering the VNIR + SWIR domain (hereafter termed VSWIR) at a resolution of 0.3 and 1.2 m, respectively, were generated and exploited in this study.

2.4.3. *Sentinel-2*

A cloud-free Sentinel-2A image covering the Bertioga forest was downloaded on the European Space Agency's Copernicus hub () (Fig. 3). The image was provided in level-2A product (*i.e.* surface reflectance) and comprised 10 spectral bands spanning over the VSWIR domain at spatial resolutions of 10 (four bands) and 20 (six bands) m. We did not use the 60-m bands because they are mainly intended for aerosol, water vapor, and cirrus applications. The image was first cropped to the extent of the site and preprocessed in ESA's SNAP toolbox (https://step.esa.int/main/download/snap-download/). All bands were resampled to two different resolutions, respectively 10 and 20 m, using the SNAP's *S2 Resampling Processor* tool (Castaldi et al., 2019), and later compared for species classification purposes.

2.4.4. *Landsat-9*

The Landsat-9 mission is the last generation of the Landsat satellite program jointly developed by the National Aeronautics and Space Administration (NASA) and the United States Geological Survey (USGS). It was launched on September 27th, 2021 (Masek et al., 2020). Landsat-9 aims to cover a broad range of applications – including vegetation mapping – by providing 30-m resolution imagery with a revisit time of eight days. The satellite embeds an Operational Land Imager (OLI) instrument covering the VSWIR range with seven spectral bands, a cirrus-specific, and a 15-m panchromatic band. In this study, a Level-2A (*i.e.* surface reflectance), cloud-free tile acquired on November 17th, 2021, and covering the entire mangrove site was downloaded on the USGS Earth Explorer platform (https://earthexplorer.usgs.gov/). The image included five spectral bands in the VNIR domain and two in the SWIR at the native spatial resolution of 30 m.

2.4.5. *DESIS*

DESIS is part of the Multi-User System for Earth Sensing (MUSES) imaging platform embedded in the International Space Station (ISS). Launched on June 29th, 2018, this instrument is operated by Teledyne Brown Engineering Inc. (Huntsville, United States) and the German Aerospace Center (DLR, Cologne, Germany). DESIS is a new generation of hyperspectral instruments that provides images at 30-m spatial resolution with a revisit time of 3 to 5 days (Kerr et al., 2016). It comprises 235 spectral bands covering the [400:1000] nm region with a resolution of 2.55 nm. The image exploited in our study was acquired on March 23th, 2022 (Fig. 3). The image was provided as a level-2A product (*i.e.* surface reflectance) on the DLR EOWEB platform (https://eoweb.dlr.de/egp/). The image was cropped to the extent of the site, and bands from the [920:960] nm region were removed because of low SNR.

2.4.6. *PRISMA*

PRISMA is a hyperspectral imaging satellite developed by the Italian

Space Agency (ASI) and launched on March 22th, 2019 (Cogliati et al., 2021). The instrument provides images that comprise 237 spectral bands covering the [400:2505] nm domain with a spectral resolution ≤ 12 nm and a spatial resolution of 30 m. PRISMA and DESIS share the same spatial resolution but differ in spectrum coverage and spectral bandwidth. A level-2D product (*i.e.* surface reflectance, geocoded image) was downloaded from the PRISMA portal. The image was acquired on July 24th, 2020 (Fig. 3) as no cloud-free image could be obtained at the time of the field campaign. However, it is unlikely that changes in the species distribution may have occurred in the Bertioga mangrove in a one-year timeframe. The image was cropped to the extent of the site and we removed the bands with low SNR as done with the airborne imagery.

2.5. Mangrove masking

Mangrove areas were isolated from their environment using the method of Baloloy et al. (2020), which proved efficient on satellite imagery. Briefly, the method consists in computing the mangrove vegetation index (MVI), defined as:

$$MVI = \frac{\rho_{NIR} - \rho_{Green}}{\rho_{SWIR} - \rho_{Green}} \quad (1)$$

where ρ_{NIR}, ρ_{SWIR}, and ρ_{Green} denote the reflectance in the Near-Infrared (NIR), SWIR, and green regions, respectively. Mangrove pixels usually exhibit higher MVI values than other vegetation and non-vegetation pixels because of a lower reflectance in the green and SWIR bands. An empirical threshold is then applied to the MVI image to isolate mangrove pixels, which are finally merged in polygons through a raster-to-vector conversion. We kept the mangrove polygon corresponding to our site, which was refined manually using known mangrove limits observed during the field survey. This way, we created a mask for each instrument, later used in the classification to avoid unnecessarily computing non-mangrove pixels. For each instrument, the wavelengths used to compute the MVI are listed in Table S2 and the main steps of the method are illustrated in supplementary Figure S2. Because the DESIS instrument does not cover the SWIR range, we used the mangrove mask generated from the PRISMA imagery as these sensors share the same spatial resolution. All mangrove masks showed high consistency with the 30-m resolution mapping of Bunting et al. (2018).

2.6. Mangrove species classification

2.6.1. Ground truth dataset generation

Using the six instruments, we trained and evaluated object- and pixel-oriented classification models at varying resolutions (0.5 to 30 m) to map mangrove species using the ground truth data collected in the field. For this purpose, we created separate Individual Tree Crown (ITC) and pixel datasets for each instrument, taking its resolution and the aim of the classification into account. We manually delineated the corresponding ITC on the airborne and 0.3-m WorldView-3 images from the tree coordinates collected in the field. Based on field observations and expert knowledge, we completed this dataset by photo-interpretation, selecting isolated trees and trees from the inner and fringe zones, and along river shorelines. This way, 714 ITCs were delineated (see Figure S3). All crown polygons were stored in a single shapefile for each instrument and later used in the object-oriented classification. For the VHR pixel-oriented classification (0.5–1.2 m), we applied a vector-to-raster conversion of the same shapefile to obtain a single image with all ITCs represented as pixels to which were assigned a unique value for each species (*e.g.* 1: Red, 2: Black, 3: White, and 0: Background).

For the 10, 20, and 30-m resolution imagery, we kept only the trees located in monospecific zones (*i.e.* surrounded by individuals of the same species) to train and assess the model on pure pixels (species unmixing being out of the scope of this study). Each ITC being encompassed in a single pixel, we substantially enlarged the dataset by adding pixels from monospecific zones to obtain a comprehensive dataset for training and testing the classification. As a result, the same dataset was used for Sentinel-2 at 10 and 20 m. Likewise, Landsat-9, DESIS, and PRISMA shared the same dataset to make classifications comparable among these instruments. The ITC and pixel count for each instrument and mangrove species are specified in Table S3.

2.6.2. Proposed classification framework

2.6.2.1. MT-EDv3 neural network. The framework developed to map mangrove species is based on the Multi-Task Encoder-Decoder (MT-EDv3) network, which was successfully applied to airborne RGB and UAV hyperspectral imagery for mapping tree species in other forest types (La Rosa et al., 2021; Martins et al., 2021). Briefly, the MT-EDv3 is a multi-task Convolutional Neural Network (CNN) that jointly performs pixel-wise regression and classification to achieve ITC delineation and species classification from a set of input images. The network involves a shared encoder based on the ResNet architecture (He et al., 2016) and two task-specific decoders. The classification-oriented decoder is based on the DeepLabv3 + architecture (Chen et al., 2018) and the regression-oriented decoder on traditional convolutions and bilinear upsampling.

The decoder designed for the regression task is used to delineate mangrove ITCs as described in Lassalle et al. (2022). It produces a normalized Euclidean distance image that denotes the distance of each pixel within ITCs to its closest crown border. This way, tree crowns are isolated from the background and can be better delineated. This decoder is trained from manually-delineated crowns (La Rosa et al., 2021). Prior to delineation, the contrast between ITC borders and out-of-crown pixels is further enhanced using a Laplacian over Gaussian (LoG) filter (Marr and Hildreth, 1980). This image filter attenuates noise in the distance image through Gaussian smoothing before finding crown borders in the second derivative of the smoothed image. Here, the LoG filter was applied with a σ of 1.5 and a kernel size of 9 (Lassalle et al., 2022). The resulting edge-enhanced distance image serves as input in a marker-controlled watershed segmentation for delineating ITCs (Huang et al., 2018; Tong et al., 2021). For this purpose, treetops are detected in the distance image using a local maximum detection filter, assuming that pixels near or at the center of crowns correspond to treetops. The detected treetops then feed the marker-controlled watershed segmentation, considering the enhanced crown borders as watershed ridge lines. The resulting polygons – which correspond to all ITCs present in the image – are then saved in a shapefile. This approach was applied to delineate crowns from the airborne and 0.3-m WorldView-3 images in the object-oriented classification.

The decoder designed for the classification task is used to map mangrove species at the pixel level. A semantic segmentation decoder learns from a set of training pixels with known species class (red, black, or white) to predict species over the entire image. Its architecture is based on atrous convolutions and pyramid scene parsing pooling. The output of the ResNet encoder feeds an Atrous Spatial Pyramid Pooling (ASPP) module, which involves parallel atrous convolutions operations and a global average pooling. The outputs of the ASPP module are concatenated and undergo traditional convolutions and a bilinear upsampling to recover the input spatial dimensions of the image. A last convolutional layer delivers the pixel-wise, per-species probability. Class membership probability is predicted for a given pixel, and the highest probable class is attributed. To achieve this, the decoder finds the weighing of input features (*i.e.* spectral bands) that best predicts classes on the training samples. The output image is a prediction of mangrove species at the pixel level and its associated probability.

2.6.2.2. Training. The MT-EDv3 was applied two ways in this study as illustrated in Fig. 4. First, the classification decoder was exploited alone to perform the pixel-oriented mapping of mangrove species, which involved all instruments described in Section 2.2.3. Besides, both

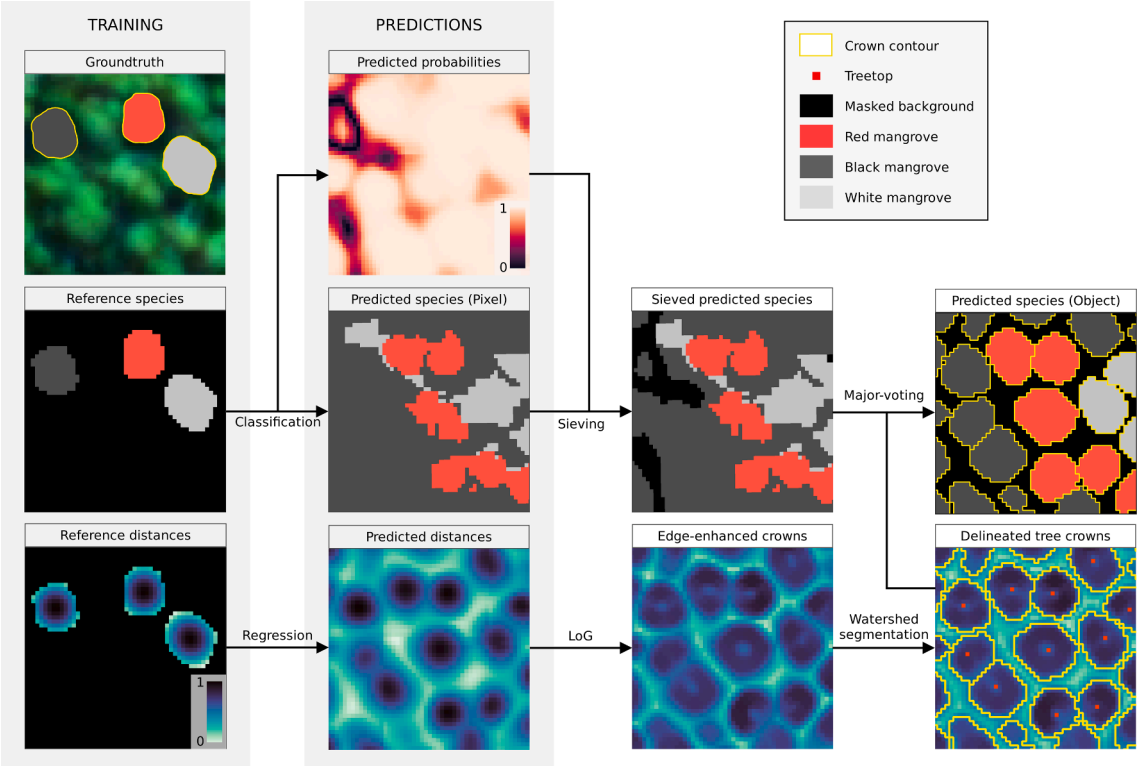


Fig. 4. Conceptual workflow of the MT-EDv3 application to map mangrove species at the pixel (Pixel) and tree crown (Object) levels. The network includes two decoders: one dedicated to pixel-oriented species classification (*i.e.* semantic segmentation) and one to pixel-oriented distance estimation (*i.e.* regression). The predicted distances are used to delineate Individual Tree Crowns (ITCs) as proposed by Lassalle et al. (2022), which are then matched with the species predictions to map mangrove species at the crown level. The "*Predicted species (Pixel)*" and "*Predicted species (Object)*" are used as final products in the pixel- and object-oriented mappings, respectively. Deep black pixels correspond to the background. (LoG: Laplacian of Gaussian.).

decoders were used jointly to perform the object-oriented mapping (*i.e.* for mapping species at the crown level) on the VHR images only (airborne and 0.3-m WorldView-3). Other images with coarser resolution (1.2 to 30 m) did not allow accurate delineation of ITCs, so they were not considered for object-oriented classification.

Three input images are needed to train the network: (i) the original reflectance image, (ii) the masked reference species classes (*i.e.* the labeled image), and (iii) the masked reference Euclidean distance image. These images were generated using two-thirds (66 %) of the ground truth pixels and ITCs (see Table S3). The train and test datasets exploited here were produced by a constrained Kennard-Stone algorithm, which guaranteed unbiased sample selection (Kennard and Stone, 1969). This algorithm first selects the two farthest samples based on their Euclidean distance and grafts subsequent samples that maximize the distance from the previously-selected ones until 66 % of the samples are selected. The remaining 33 % form the test dataset. This way, representative train and test datasets are built. To ensure total independence of train and test samples on VHR imagery, we constrained the Kennard-Stone algorithm so that when it selects a crown pixel, all pixels from that crown are assigned to the same dataset. Consequently, none of the pixels used for assessing the accuracy of species predictions served to train the classification models, thus preventing the influence of spatial autocorrelation.

The MT-EDv3 network was trained on the partial loss principle, which relies on computing a loss function from the annotated pixels, thus skipping pixels without annotations (*i.e.* background) (Lin et al., 2017). This approach is performs well using a limited set of annotated samples (La Rosa et al., 2021; Martins et al., 2021). The MT-EDv3 network is trained by generating N image patches of $n \times n$ pixels, using 90 % of them for training and 10 % for parameter validation and verifying that at least 10 % of pixels within the patch include a reference sample. The number of patches (N) and the patch size ($n \times n$) were defined according to the spatial resolution of the images. They are specified for each instrument in Table S4. We implemented data augmentation in the training process by randomly rotating (90, 180, and 270°) and flipping image patches to enhance network generalization. Overfitting was safely alleviated by setting a dropout rate of 0.65 and we used a stochastic gradient descent optimizer with a momentum of 0.9 and initial learning of 0.05 (Murphy, 2012). The MT-EDv3 network was trained for maximal 20 epochs and a batch size of 8. The training process automatically stopped if no increase in F1-score was observed in the validation patches after 5 consecutive epochs. Each epoch exploited a new set of training and validation patches, and final predictions were made using a sliding window of 10, 30, and 50 % overlapping rates and by computing the mean over the three outputs. A weighted categorical focal loss function was implemented so that class weights were determined inversely to the proportion of samples (Lin et al., 2017).

2.6.2.3. Class attribution over the entire image. The trained model was applied to predict the distances and classes over the entire reflectance images and ITCs were automatically delineated as described in section 2.3.1. The class prediction image was the final product of the pixel-oriented species mappings. Regarding object-oriented mappings, the output ITC polygons obtained from the watershed segmentation were matched with the predicted pixel classes of the classification decoder. That is, species classes were attributed to each crown based on the predictions of the corresponding pixels. As pixels from a single crown may differ in the predicted species classes, we applied the following two-step procedure to label ITCs. First, pixels for which the highest-class probability fell below 0.7 were not considered as they corresponded to low-confidence predictions. Then, the species class of each crown polygon was determined using a majority-voting rule (Aval et al., 2019). The same weight would be given to low- and high-confidence

predictions without this rule, leading to inaccurate species labeling. In the case of a draw (*e.g.* 50 % of red mangrove and 50 % of black mangrove pixels), the class with the highest average probability over the entire tree crown was retained.

The MT-EDv3 network was applied separately for each approach (pixel *vs* object), instrument, and spatial resolution, using either the VNIR or the VSWIR bands to determine the influence of the spatial and spectral resolutions, and spectrum range, on species separability. So, 24 mapping experiments were carried out, on a computer equipped with two Intel® Xeon Silver processors (2.40 GHz), 128 GB RAM, and an 8-GB NVIDIA® RTX 2070 SUPER graphic card. All codes were developed in Python using TensorFlow and CUDA® 11.4 and can be made available by authors upon request.

2.6.3. Assessment and comparison with other classifiers

Species classification was assessed by comparing the actual mapping to predictions obtained with the MT-EDv3 network, on the test dataset (33 % of ground truth data), which did not serve for training. This was achieved through a pixel-to-pixel comparison for the pixel-oriented classification, while a best-matching, crown-to-crown comparison was applied for the object-oriented classification. Each actual ITC polygon was compared to the best-overlapping predicted ITC, considering a minimum Intersection over Union (IoU) threshold of 50 % (Tochon et al., 2015). For each experiment, predictions were evaluated using the Overall Accuracy (OA), Cohen's Kappa, and the F1-score of each species computed from the confusion matrices (Cohen, 1968; Ferreira et al., 2016). For concision, results obtained with WorldView and Sentinel-2 are illustrated for the most satisfactory resolution – respectively 0.3 and 10 m – compared to those obtained with the coarser resolution products – respectively 1.2 and 20 m. For 30-m hyperspectral spaceborne imagery, we used bands from the DESIS images for VNIR-based classification, whereas PRISMA bands were exploited in the VSWIR-based classification. Results obtained with DESIS (VNIR) and PRISMA (VSWIR) imagery were jointly compared.

The results obtained with the MT-EDv3 network were compared to those of commonly-used classifiers, which successfully mapped mangrove species in previous studies. This included Support Vector Machine (SVM) with a radial basis function (Pham and Brabyn, 2017; Wan et al., 2018), Random Forest (Bullock et al., 2017), and Maximum Likelihood Classification (MLC) (Vaiphasa et al., 2006; Wang et al., 2004). These classifiers were applied following the same training and test scheme as that of the MT-EDv3 network (66 % training, 33 % testing), and SVM and Random Forest parameters (C, γ, number of trees, *etc*.) were optimized using a 5-fold cross-validation strategy (Hastie et al., 2009). The resulting pixel- and object-oriented mappings were compared to the ground truth using the metrics described in section 2.3.3. Because SVM, Random Forest, and MLC do not perform ITC delineation, we used the ITC polygons obtained with the MT-EDv3 network for the object-oriented classification. Still, we matched them with the species maps obtained by these classifiers at the pixel level. In other words, we replaced the output of the MT-EDv3 classification task with that of the SVM, Random Forest, or MLC to obtain the object-oriented mapping of species.

3. Results and discussion

3.1. Advances in mangrove species mapping

Based on our review, most of previous studies (80 %) focused on developing spaceborne-based approaches to map mangrove species due to the availability of free and commercial image products. According to those studies, the classification accuracy does not seem affected by the number of species in a mangrove stand (see Figure S4). This is also true for airborne and UAV imagery. However, the accuracy is notably determined by the type of image used in classification models. Historically, QuickBird, IKONOS, Pleiades, Landsat, and SPOT images were exploited to map mangrove species in the early 2000 s, reaching on average 82 % and 0.77 of OA and Kappa coefficient, respectively (Aschbacher et al., 1995; Neukermans et al., 2008; Ramírez-García et al., 1998; Viennois et al., 2016; Wang et al., 2004) (see Figure S5). Most studies published after 2015 were based on Sentinel-2 images, thus benefiting from a broad spectrum coverage and narrower bandwidth but limited to medium spatial resolution (10–60 m) (Manna and Raychaudhuri, 2020; Valderrama-Landeros et al., 2018; Wang et al., 2018; Xia et al., 2020). No or minor improvements were reported (OA = 79 % and Kappa = 0.72) with the addition of the Short-Wave Infrared (SWIR, 1000–2400 nm) bands, which highlighted the limitations of medium- and high-resolution multispectral imagery in mapping mangrove species. From that point, two solutions were adopted to improve species classification. First, a few studies opted for VHR multispectral imagery such as WorldView-2 and −3, sometimes by further enhancing the resolution through pan-sharpening (Li et al., 2021; Viennois et al., 2016; Wan et al., 2019, 2018). VHR imagery greatly improved species classification as studies reported OA and Kappa values averaging 86 % and 0.82, respectively. On the other hand, some studies showed that hyperspectral, medium-spatial resolution imagery acquired by Hyperion and GeoFen-5 satellites provide comparable classification performances to VHR multispectral imagery (average OA = 88 %, Kappa = 0.84) (Kumar et al., 2019; Wan et al., 2020; Wong and Fung, 2014). Thus, both strategies (*i.e.* VHR multispectral and medium-resolution hyperspectral imagery) proved successful in improving mangrove species mapping, suggesting that higher spatial resolution can alleviate the need for hyperspectral imagery, and conversely. Promising applications are expected with the newly-operating hyperspectral spaceborne instruments. For example, DESIS and PRISMA instruments show great potential for mapping mangrove species but were never used for this purpose before the current study.

Less than 20 % of articles published until the end of 2021 involved airborne and UAV imagery. However, they all highlighted the importance of very-high spatial and spectral resolutions. Airborne imagery achieves an average OA of 83 % in discriminating among mangrove species at 0.5–4 m spatial resolution, with no apparent differences between multi- and hyperspectral instruments (see Figure S5) (Chaube et al., 2019; Li et al., 2021). Thus, there is no evidence that airborne imagery outperforms spaceborne imagery despite higher spatial and spectral resolutions (Osei Darko et al., 2021; Prakash Hati et al., 2021). More recently, compact cameras designed for UAVs have offered new, cost-effective applications in mangrove monitoring (Zimudzi et al., 2021). UAVs greatly complement spaceborne imagery in mangrove conservation as they allow on-demand image acquisitions. As mangroves establish mainly in tropical and neotropical regions, obtaining cloudless satellite imagery can be tricky even with a daily-revisit time. UAVs can be deployed as soon as clear sky conditions are met, filling data gaps caused by unsuccessful satellite acquisitions. Moreover, they allow ultra-fine monitoring thanks to cm-scale resolution. Up to date, only seven studies have reported the use of UAV imagery for mapping mangrove species, with classification models averaging OA and Kappa of 88 % and 0.85, respectively (see Figure S5) (Cao et al., 2021, 2018; Chaube et al., 2019; He et al., 2020; Jiang et al., 2021; Z. Li et al., 2019; Zhu et al., 2019). Four studies exploited VNIR hyperspectral imagery (138 to 226 bands) and reported slightly better performances than multispectral cameras (see Figure S5). However, none of these studies considered the SWIR relevant to improving species discrimination. Thus, little is known about the capability of UAVs to monitor mangrove diversity, and extensive efforts are still required to make them reliable for operational applications.

Of all studies published until 2021, more than 70 % adopted pixel-oriented approaches to map mangrove species, regardless of the sensor type (UAV, airborne, spaceborne). This is because object-oriented classification (i) is almost exclusive to VHR imagery (less than20 % of studies) and (ii) involves the implementation of image segmentation before classification. Pixel-oriented classification is a more direct

approach as no prior segmentation is needed and can be applied at any spatial resolution. Object-oriented classification allows mapping species at the ITC level, which shows many advantages in forest management, monitoring, and conservation (Bunting and Lucas, 2006; Dalponte et al., 2014; Ferreira et al., 2016). There is no clear evidence on the best approach to classify mangrove species (see Figure S6). Studies remain poorly comparable as they differ in study sites, species classes, imagery products, and classifiers. Yet ITC delineation remains highly challenging in mangroves because of crown clumping and understory vegetation (Lassalle et al., 2022). In previous studies, object-oriented approaches involved grouping trees in segments rather than delineating tree crowns individually (Cao et al., 2021; Kamal and Phinn, 2011; Pham and Brabyn, 2017; Zhu et al., 2020). Thus, no approach has been proposed to map mangrove species accurately at the crown level.

3.2. Proposed species classification framework

3.2.1. Performances on VHR imagery

The results of OA and Kappa coefficient obtained with the proposed framework at pixel and object levels on VHR imagery are presented in Fig. 5 and detailed in Table S5. VHR imagery outperformed high- and medium- resolution imagery with OA and Kappa coefficients exceeding 80 % and 0.75, respectively, which is consistent with our observations from the literature (see Figure S4). Comparable performances were observed between airborne and pan-sharpened WorldView-3 imagery (0.3 m) at both object and pixel levels. More generally, no major differences were noticed between pixel- and object-oriented mappings for these instruments, highlighting our approach's capability to perform well for both classifications. This observation reveals that, at very fine resolution, hyperspectral imagery is not mandatory to achieve accurate mapping of mangrove species. It also confirmed the trend observed in the literature, according to which mangrove species are equally distinguishable at pixel and object levels (see Figure S6).

Regarding WorldView-3, pan-sharpening improved the pixel-based classification compared to the native 1.2-m resolution. Also, adding the SWIR bands drastically improved species classification as OA and Kappa increased by 8 % and 0.11, respectively. The same observation was made on airborne hyperspectral images to a lesser extent. SWIR bands had less influence on the object-oriented classification as a few changes in pixel labeling within a single crown had little influence on the majority-voting decision. Most inaccuracies in species classification came from confusion between the red and black mangroves. Based on the analysis of F1-scores (Fig. 6), the white mangrove was almost perfectly distinguished at all scales, even when only the VNIR bands were used. Only slight improvements were observed in the discrimination of this species when adding the SWIR bands. Thus, the main benefits of these bands relied on improving the discrimination of red and black mangroves as shown by higher F1-scores for these species. When using the entire spectrum, confusions were reduced by up to 81 and 61 % at pixel and object levels, respectively, for these species (see confusion matrices in supplementary Tables S6-S7). SWIR bands were significant on WorldView-3 imagery, which showed the highest increase in F1-scores.

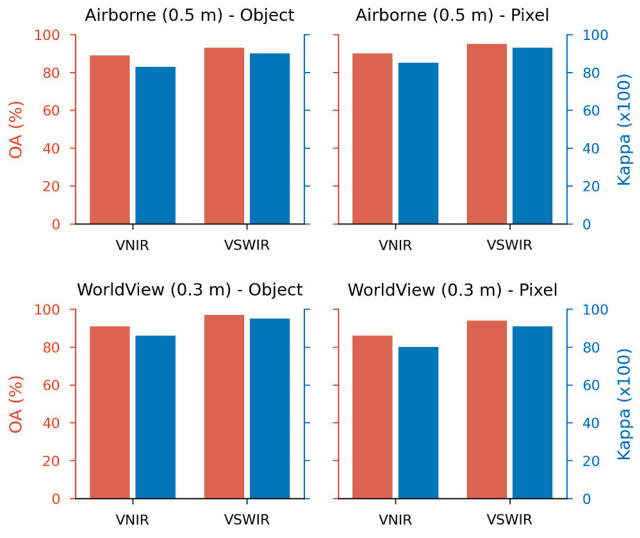


Fig. 5. Overall Accuracy (OA) and Kappa coefficients of the species classification obtained with the very-high-resolution instruments at the object (*i.e.* individual tree crown) and pixel levels using either the VNIR or VSWIR bands. Red and blue bars depict the OA (left axis) and Kappa (right axis), respectively. Kappa values were multiplied by 100 for graphical convenience. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2.2. Performances on high and medium-resolution imagery

The OA and Kappa obtained with the high- and medium-resolution images are presented in Fig. 7. Although performances obtained at 10–30 m resolution did not equal those of VHR images, high accuracy was observed using VSWIR bands for Sentinel-2 at 10-m resolution and PRISMA imagery at 30 m (OA $\geq$ 88, Kappa $\geq$ 0.80). Likewise, promising results were obtained on DESIS imagery (OA = 85, Kappa = 0.75) despite the instrument covering only the VNIR. Conversely, weaker performances obtained with Sentinel-2 at 20-m resolution and Landsat-9 (OA $\leq$ 82, Kappa $\leq$ 0.73) emphasized the limits of multispectral imagery to map mangrove species at medium spatial resolution. Once again, confusion between the red and black mangroves was reduced by adding SWIR bands (up to 54 %), especially on Sentinel-2 imagery (Fig. 8, see also Tables S6-S7).

3.2.3. Comparison with other classifiers

Historically, traditional and machine learning classifiers have been widely used for mangrove species mapping. Our review shows that Spectral Angle Mapper (SAM) and MLC were exploited in 9 and 16 % of studies, respectively. These classifiers were successfully applied to multispectral imagery before 2015 and still compete with new algorithms (Kripa et al., 2020; Pandey et al., 2019; Wang et al., 2004). They

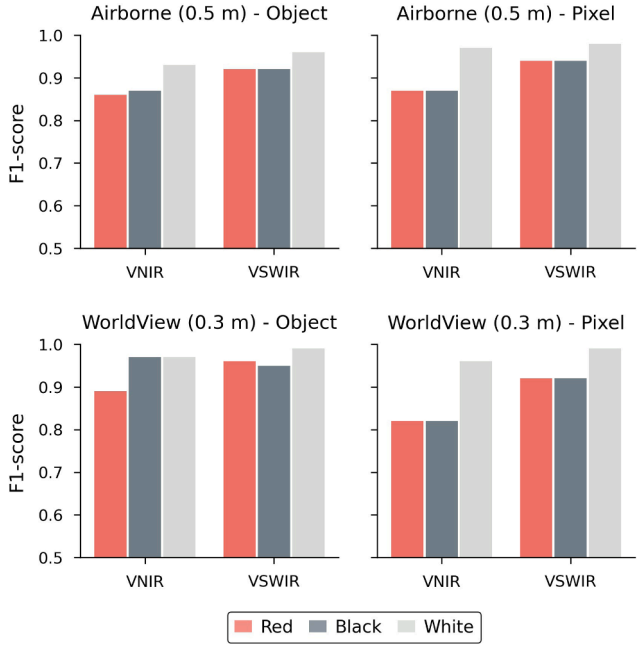


Fig. 6. F1-scores obtained for the classified mangrove species with the very-high-resolution instruments at the object (*i.e.* individual tree crown) and pixel levels using either the VNIR or VSWIR bands.

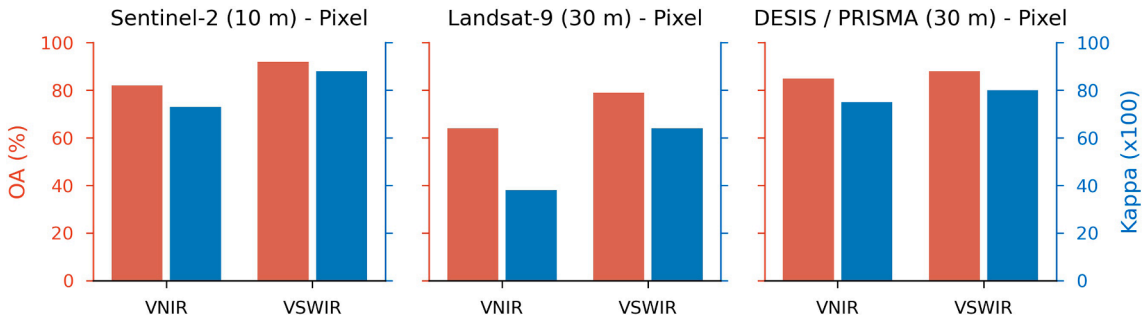


Fig. 7. Overall Accuracy (OA) and Kappa coefficients of the species classification obtained with the high- and medium-resolution spaceborne instruments at the pixel level using either the VNIR or VSWIR bands. Red and blue bars depict the OA (left axis) and Kappa (right axis), respectively. Kappa values were multiplied by 100 for graphical convenience. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

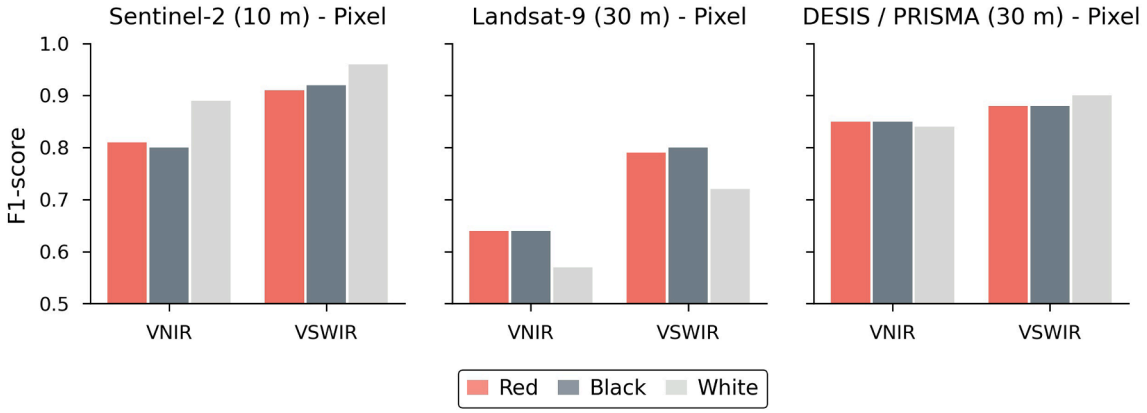


Fig. 8. F1-scores obtained for the classified mangrove species with the high- and medium-resolution spaceborne instruments at the pixel level using either the VNIR or VSWIR bands.

become, however, inappropriate for use on hyperspectral imagery as they poorly handle band collinearity and class imbalance. So, SAM and MLC classifiers have been progressively replaced by machine learning algorithms to cope with the high dimensionality of hyperspectral data and, sometimes, the limited number of training pixels in classification problems. More than 40 % of approaches developed until 2021 exploited either SVM or Random Forest to map mangrove species, achieving an average OA of 84 % (Bullock et al., 2017; Jiang et al., 2021; Pham and Brabyn, 2017; Rahman et al., 2019; Zhu et al., 2020). A few implemented band selection and dimensionality reduction, but the benefits to species mapping remain poorly documented. Yet some limitations apply to machine learning algorithms in mangrove studies. First, they are designed to achieve a single task. So, machine learning algorithms cannot solve segmentation and classification problems simultaneously, making them unsuited for object-oriented mapping. In addition, machine learning showed poor transferability among instruments with varying spatial and spectral resolutions and require feature engineering by the user. Deep learning could overcome those limitations, which proved a powerful alternative to machine learning in vegetation studies over the past two decades (Kattenborn et al., 2021). In particular, multi-task CNNs are given increasing attention in species mapping as they achieve accurate delineation of tree crowns and species classification simultaneously, simplifying object-oriented mapping (Chiang et al., 2020; Zhu et al., 2017). Moreover, CNNs are not site- and instrument-specific, showing good transferability for global mangrove management. These networks have been successfully applied to various forest types, including urban vegetation and tropical forests (Ferreira et al., 2020; La Rosa et al., 2021; Martins et al., 2021). However, no application has been proposed for mapping mangrove species as previous studies exploited single-task networks with simplified architectures to perform pixel-oriented classification (Held et al., 2003; Wang et al., 2008; Wong and Fung, 2014). Our approach is, therefore, the first to exploit a multi-task network for this purpose. According to the performance comparison, our framework outperformed other classifiers at pixel and object levels (see the detailed OA and Kappa in supplementary Table S8). OA was 12, 14, and 24 % higher using the MT-EDv3 network than SVM, Random Forest, and MLC, respectively. These results are consistent with a recent study involving a similar classifier comparison (Li et al., 2021). MLC exhibited the poorest performance, especially on hyperspectral imagery as this classifier poorly handles high dimensionality and collinearity among bands. Hence, MLC shows poor generalization across instruments and should be avoided when using images with hundreds of spectral bands. Although SVM and Random Forest performed well, their application to hyperspectral imagery – especially DESIS and PRISMA – was sometimes limited by unbalance between the number of training samples and input features. On very few occasions, SVM led to better results than the MT-EDv3 network, but differences in OA never exceeded 5 % and only involved Sentinel-2 (10-m) and Landsat-9 imagery. Moreover, one major advantage of our framework lies in its capability to perform all steps required to achieve object-oriented classification (*i.e.* ITC delineation and species labeling) on its own. This is not the case of other approaches, which rely on prior delineation of ITCs using a dedicated approach. Our framework can achieve accurate pixel- and object-oriented classification over a broad range of optical instruments, including recent ones (Landsat-9, DESIS, PRISMA). The MT-EDv3 network is easy to parametrize and requires no preprocessing of original images except resampling or pan-sharpening, but these steps might be skipped. It requires, however, a large dataset of independent, training ITCs to take the intraspecific variability of reflectance and crown characteristics into account. Recent improvements opened the way to applications on limited datasets by implementing a new partial loss function that achieves accurate semantic

labeling from a low number of training samples (La Rosa et al., 2021).

3.2.4. Application over the entire study site

The maps of mangrove species produced over the entire study site using the proposed approach are presented at the pixel level in Fig. 9. Species zonation patterns were consistent among sensors, field observations, and known species segregation. Red mangroves mainly border river shorelines and the forest's seaward segment, which are directly exposed to tide and under permanent flooding. According to pixel-oriented maps, this species covered 35–47 % of the forest (Fig. 10). Black mangroves are found in the inner areas, sometimes mixed with red mangroves (only visible on VHR imagery), occupying 40–57 % of the forest area. Finally, white mangroves are almost only found in the innermost segment, close to the forest limit, which is only flooded by freshwater during the rainy season. Some individuals are close to the tide-exposed limit, but none are exposed to permanent saline flooding. White mangroves represent only 7–10 % of the forest area.

According to our review and based on these results, 30-m multispectral imagery should be used with caution when mapping mangrove species (Valderrama-Landeros et al., 2018; Wan et al., 2020; Wang et al., 2018). For example, Landsat-9 respected the overall zonation but was unable to catch local variations in species distribution. Conversely, 10-m Sentinel-2 imagery showed excellent performance using the VSWIR, highlighting that SWIR bands (1610 and 2190 nm) benefit species discrimination. The results are consistent with previous studies (Manna and Raychaudhuri, 2020; Prakash Hati et al., 2021; Qiu et al., 2019; Valderrama-Landeros et al., 2018). One advantage of using Sentinel-2 images at 10-m over 20-m resolution is detecting islets of trees outside their main species zone, which provides a more refined mapping of species distribution (Fig. 9). However, the benefits of SWIR bands were limited at 10-m resolution because they were resampled from their native resolution (20 m). The same applies to WorldView-3 imagery, of which 3.7 m SWIR bands were pan-sharpened to 0.3 m.

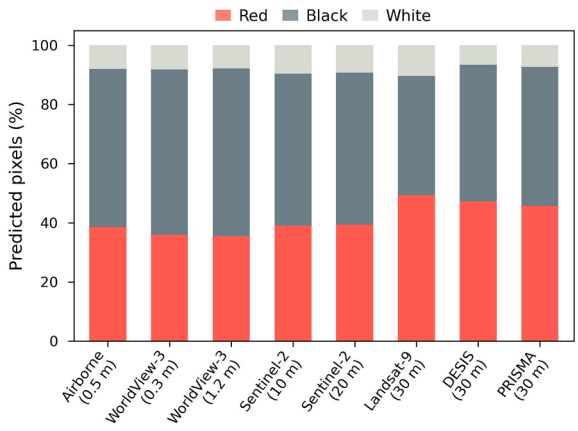


Fig. 10. Relative proportion of red, black, and white mangroves over the entire Bertioga mangrove according to pixel-oriented mappings obtained with various sensors. The spatial distribution of species is illustrated in Fig. 9. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Regarding hyperspectral instruments, Hyperion and Gaofen-5 opened the way to mapping mangrove species from space using hyperspectral imagery, achieving reasonable mappings (Koedsin and Vaiphasa, 2013; Wan et al., 2020). DESIS and PRISMA, which offer narrower bandwidth and higher SNR, allow better distinction of spectrally-similar species, reducing confusion. A comprehensive comparison of all these instruments would be needed to confirm this assumption. Nevertheless, our study shows that broad spectrum coverage (PRISMA) is more beneficial than narrower bandwidth and higher SNR (DESIS) as the SWIR brings non-redundant information with the VNIR in species discrimination. While reflectance in the VNIR is

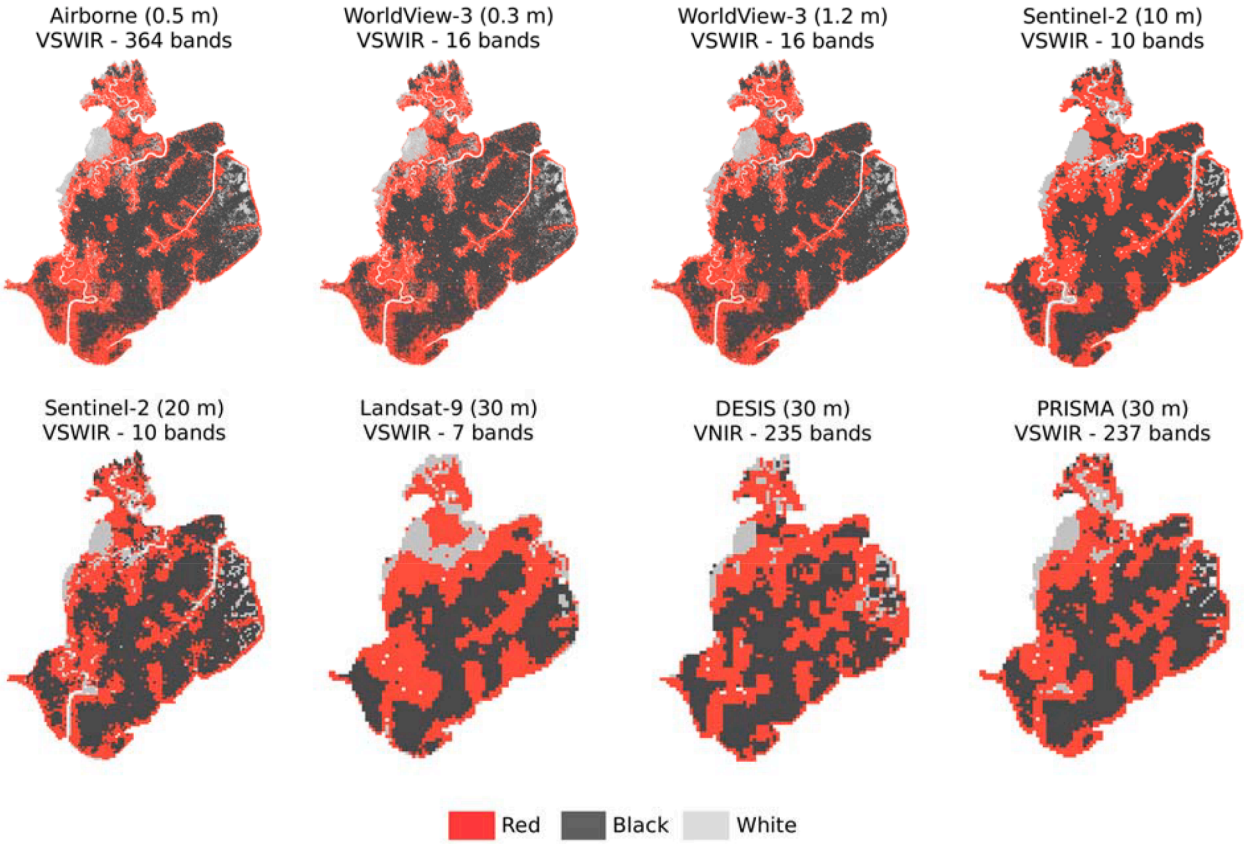


Fig. 9. Pixel-oriented mapping of mangrove species obtained from the multi- and hyperspectral images at varying spatial resolutions using the proposed framework.

mainly driven by leaf pigment (chlorophylls, carotenoids, and anthocyanins), water, and dry matter contents, reflectance in the SWIR is linked to carbon-based constituents (celluloses, lignins, sugars), protein, and also water contents (Féret et al., 2021, 2017). So, if species differ in their biochemistry, they will also differ in reflectance but not in the same way in the VNIR and SWIR. This may explain the additional information brought by the SWIR bands and thus the improvements observed in species classification here. As the 30-m resolution is limited, PRISMA also offers a 5-m resolution panchromatic channel, which opens the way to hyperspectral pan-sharpening. This may overcome limitations observed at the native 30-m resolution.

Subsets of the object-oriented maps obtained from airborne (0.5 m) and pan-sharpened WorldView-3 imagery (0.3 m) are provided in Fig. 11. ITC delineation was successfully achieved as the crown area (12.6 m^2 on average, see Lassalle et al. (2022)) was much larger than pixel size, so crowns were accurately captured using our framework, regardless of their size and shape. High consistency was observed among the three scales, although finer spatial resolution slightly improved the delineation as observed visually between airborne and WorldView-3 images (Fig. 11). Therefore, our framework is the first to achieve a "true" mapping of mangrove species at the tree crown level. This is because the classification obtained with the MT-EDv3 network benefited from the high accuracy of ITC delineation, limiting the inclusion of out-of-crown pixels into crown polygons. This delineation task was formerly developed for 0.3-m WorldView imagery and was successfully adapted to 0.5-m airborne hyperspectral imagery here. This was possible because 0.5-m pixels are lower than the average crown diameter observed in this forest (Soares and Schaeffer-Novelli, 2005) (Fig. 11). As a general rule, it is considered that pixel size must be smaller than the crown diameter to achieve accurate delineation (Ke and Quackenbush, 2011). In our case, coarser-resolution such as the native 1.2-m VNIR bands of WorldView-3 would result in grouping crowns from different species into the same segment, introducing confusion in the classification. In several mangroves, however, the average crown diameter exceeds 10 m (Wannasiri et al., 2013), allowing ITC delineation without prior pan-sharpening. Since species are poorly mixed within our study site, both pixel- and object-oriented mappings agreed with ground truth data. In mangroves with higher heterogeneity, the object-oriented classification might lead to better results, as coupling a species probability threshold to a majority-voting rule limits the influence of local heterogeneity on species predictions. Also, refining this threshold may improve the predictions. Species labeling could be further improved by considering the crown attributes (*e.g.* area, eccentricity, compactness). Such features can be easily estimated from segmented ITC polygons and exploited as an additional parameter in species labeling.

3.3. Perspectives towards improving mangrove species mapping

Our framework overcomes several limitations from previous studies, by ensuring good sensor transferability and accurate ITC-oriented mapping through multi-tasking (delineation and classification). The simple context as studied here (three species) is a necessary step to unravel the influence of instrument specifications and species diversity on classification accuracy. As a next step, it would be necessary to test our framework on other sites with higher species richness and spatial heterogeneity to evaluate its limitation in regard to multi- and hyperspectral instruments, and to adapt it accordingly. Our review identified these two factors as among the most limiting to classification performances, especially on spaceborne imagery. As the number of species and the spatial heterogeneity of the forest increase, species become more mixed in pixels and decreasingly separable spectrally. In highly-diversified mangroves, species can be mixed both horizontally and vertically so that only VHR hyperspectral imagery provides satisfying accuracy (Chaube et al., 2019; Li et al., 2021). Comparing airborne hyperspectral imagery to Hyperion, Sentinel-2, and Landsat-8, Prakash Hati et al. (2021) showed that only high spatial-spectral resolutions allow for identifying species that are too sparse to be identified at coarser resolutions. Consequently, spaceborne imagery may also be limited in mangroves with high spatial heterogeneity. In addition, environmental stressors can impact tree health differentially among mangrove species, affecting the spectral variability within and among species and, consequently, species separability from spaceborne imagery (Flores-de-Santiago et al., 2016; Zhang et al., 2014). Hence, although our framework showed promising capability to map species from space in a low-diversity mangrove, significant improvements would be necessary to extend its scope to other contexts. In that respect, three perspectives are discussed hereafter: (i) spectral unmixing, (ii) multi-temporal imagery, and (iii) active and passive data fusion.

Hyperspectral imagery opened the way for spectral unmixing at medium resolution, which allows for identifying the abundance of mixed species in pixels (Heylen et al., 2014; Quintano et al., 2012). Yet, only a few studies (less than5%) have applied spectral unmixing to map mangrove species from spaceborne hyperspectral imagery, leading to relatively good accuracy (Chakravortty and Chakrabarti, 2017; Salghuna and Pillutla, 2017). Only Hyperion imagery was used for this purpose. DESIS and PRISMA are promising alternatives in highly-

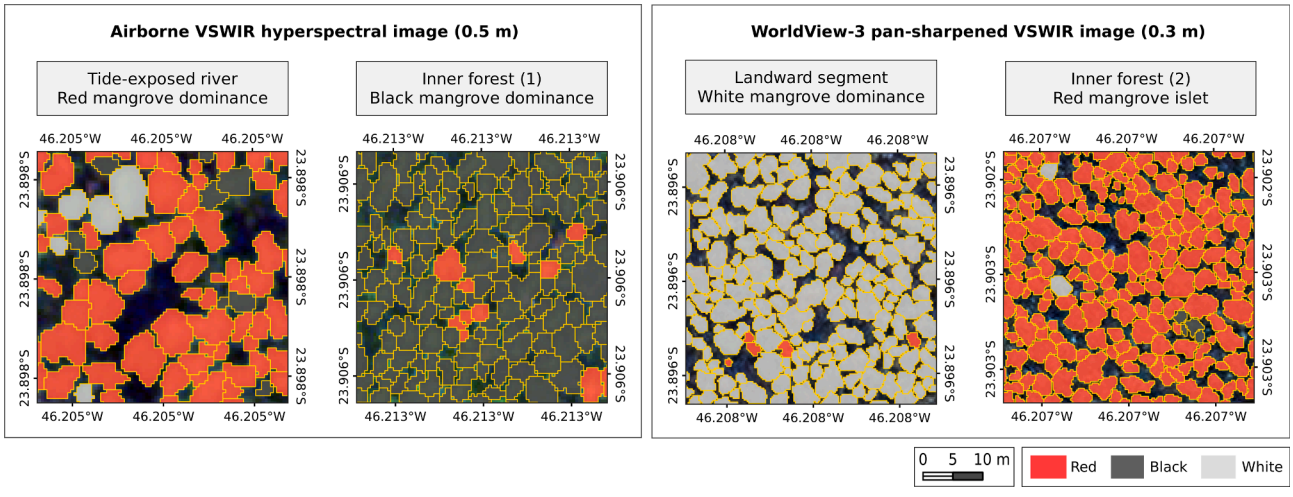


Fig. 11. Subsets of object-oriented mappings of mangrove species predicted at airborne and spaceborne scales using the proposed framework. Individual Tree Crown (ITC) polygons were automatically delineated and matched with the pixel-oriented classification using a majority-voting rule. ITC contours are marked in yellow. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

diversified, spatially-heterogeneous mangroves thanks to higher SNR and narrower bandwidth. So, spectral unmixing could be implemented in our framework to predict species abundances in pixels at medium spatial resolution using spaceborne hyperspectral imagery. Besides, using multitemporal imagery as input of the MT-EDv3 network could significantly improve the discrimination of species as they differ in their phenology (Flores-de-Santiago et al., 2016; Valderrama-Landeros et al., 2018). According to our synthesis, less than 10 % of studies have exploited multi-date imagery in classification models, but encouraging results have been reported. For example, Valderrama-Landeros et al. (2018) analyzed time-series of vegetation indices to determine the optimal timing of Sentinel-2 images to discriminate among red, black, and white mangroves, increasing OA by 24 %. Multitemporal imagery could also help to cope with environmental stressors as species are not affected in the same way by stressors over time (Flores-de-Santiago et al., 2016; Zhang et al., 2014). Combining active remote sensing data with passive optical imagery is another promising way to improve the classification of mangrove species using our framework. Data fusion remains underestimated as about 8 % of studies have considered combining LiDAR or Synthetic Aperture Radar (SAR) with multi- or hyperspectral imagery (Cao et al., 2021; Li et al., 2021; Wang et al., 2022; Wong and Fung, 2014). SAR data acquired in L, C, and P bands penetrate upper canopies and provide information about tree morphology and architecture by exploiting surface, double bounce, and volume scattering effects (Fernandez-Ordonez et al., 2009; Wang et al., 2019). As mangrove species are structurally diverse, SAR backscattering may improve species classification significantly. Based on this assumption, Rahman et al. (2019) used TanDEM-X satellite SAR data to improve species discrimination obtained with WorldView-2 imagery in medium-diversity mangroves. Also, Sentinel-1 (C-band), showed promising potential in mapping mangrove species combined with Sentinel-2 imagery (Xia et al., 2020). Satellite hyperspectral imagery has been successfully combined with Envisat ASAR C-band data for the same purpose (Wong and Fung, 2014). Alternatively, LiDAR-derived features can complement passive optical imagery by informing on canopy height, vertical architecture, and lower vegetation strata (Cao et al., 2021; Li et al., 2021; Q. Li *et al.*, 2019; Yin and Wang, 2019). The benefits of LiDAR in mangroves depend on the point cloud density and the laser beam penetration through the canopy. Combining WorldView-3 or airborne hyperspectral imagery with LiDAR at a density of 20 pts.m^{-2} significantly improved species classification in multi-strata mangroves (Li et al., 2021; Q. Li *et al.*, 2019). At higher density (91 pts.m^{-2}), LiDAR-derived canopy height models also helped to delineate ITC and could significantly improve the object-oriented classification performed with our framework. This improvement is necessary in dense mangroves with crown clumping and underlying vegetation, where ITC delineation based on passive optical imagery remains difficult (Lassalle et al., 2022).

4. Conclusion

The objective of this study was twofold. First, we summarized the advances in mangrove species mapping. Second, we proposed a multi-scale framework using multi- and hyperspectral imagery. Our framework based on a multitask CNN accurately classified mangrove species at both pixel and ITC levels using airborne and freely or commercial spaceborne imagery. The performances of our framework greatly varied according to the spatial and spectral resolutions of instruments and open up promising perspectives in the way of mangrove species mapping. In particular, we highlighted the potential of using multitemporal imagery and coupling active and passive data to improve species classification in highly-diversified mangroves. These improvements could be implemented in our framework towards extending its use to other mangrove forests. Spaceborne hyperspectral imagery gets the wind in its sails owing to recent and upcoming missions such as EnMAP, EMIT, SBG, CHIME, BIODIVERSITY, and HISUI (Cawse-Nicholson et al., 2021; Connelly et al., 2021; Loizzo et al., 2019; Matsunaga et al., 2018; Michel et al., 2011). These instruments will increase the availability of hyperspectral images over mangroves, so their performance should be benchmarked in future studies to determine the optimal specifications to best map mangrove species. Mangrove forests are subject to ambitious conservation and restoration goals globally (Buelow et al., 2022; Spalding and Leal, 2021). Remote sensing could help to achieve these goals, but substantial efforts are still required to evaluate the applications of new and upcoming instruments in mangroves. So, as a perspective of this study, we consider integrating our framework into a broader scope towards monitoring tree stress at the species level, which remains an overlooked aspect of remote sensing of mangroves (Wang et al., 2019).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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